Asking for the Whole Plan: Prompt Design Moves Language Models' Scores on a Clinical-Safety Benchmark More Than Model Choice

An exploratory study on the NOHARM v2 public cases

Ahmad Omar Aljamal, MD
Independent researcher, United States
ORCID 0000-0003-1630-1468 Correspondence: omar@aljamal.work

Version 2. This version replaces version 1 (posted October 16, 2026). The physician review was repeated item by item; its results replace the overall verdict reported in version 1, which found no added option harmful (see *Safety of the added content*). The text is condensed, with secondary analyses in the Supplementary Appendix, and spelling is American. No model output, AI evaluation or benchmark score changed.

Abstract

Objective: To measure how much a structured instruction changes large language models' scores on NOHARM, a specialist-consultation benchmark graded mainly on omissions, which part of the instruction carries the change, and whether the added content is safe.

Materials and Methods: Twelve models answered the 30 public NOHARM v2 cases (330 items) uninstructed, and 5 also with an instruction requesting a complete, structured plan. An AI evaluator applied the benchmark's severity-weighted F1. On 2 models, each part of the instruction was tested alone and against a request for 25 recommendations. Model raters and a physician reviewed the added actions.

Results: The instruction raised F1 by 0.14 on average [0.09, 0.20] and improved 4 of 5 models, most for the tersest (Grok 4.7, 0.354 to 0.633; GPT-5.6 Terra, 0.678 to 0.853). Three instructed models exceeded the best uninstructed score (0.764). The standard deviation between models halved (0.17 to 0.09). The plan format alone matched the full instruction; the framing alone or 25 recommendations scored lower. Most recovered actions were follow-up and counseling. One model proposed +0.13 [+0.05, +0.23] more specialist-rated harmful options per item; a physician judged 13 of the 43 added options harmful as stated.

Discussion: On this benchmark, an uninstructed score reflects how much a model volunteers, not only what it knows. A structured instruction recovered omitted care but added some harmful options.

Conclusion: Benchmarks of clinical completeness should report scores under a standard structured instruction beside uninstructed scores, and check what the instruction adds, before models or clinical tools are ranked.

Keywords: large language models; decision support systems, clinical; patient safety; benchmarking; referral and consultation

Background and Significance

NOHARM is a benchmark of the safety of language-model recommendations in specialist consultation.[1] It compares a model's management plan for a real primary-care-to-specialist electronic consultation with specialists' ratings of the actions the case calls for and those it should avoid, and weights each error by the harm it could cause. In its evaluation, most severe errors were omissions, and recall trailed precision for nearly every model. The public leaderboard scores each model on the bare case, without an instruction.

How a model is prompted is known to affect its benchmark scores.[2–6] NOHARM's authors found that priming prompts asking for completeness, capped at 200 or 500 words, improved the scores of generalist models, although clinical retrieval tools remained ahead.[1] If coverage depends on how a model is asked, a bare score reflects how much a model volunteers as well as what it knows, and a ranking built on bare scores could mislead anyone choosing a model or a tool. However, a gain from a longer answer is difficult to interpret. AI evaluators favor longer and self-generated text,[7–11] although a length-matched analysis on HealthBench found this effect to be partial,[12] and evaluators, like models, detect missing content poorly.[13,14] A gain from an instruction must therefore be separated from a gain from length, and the content it adds must be checked for harm that the rubric cannot detect.

Objective

We sought to evaluate how much a structured instruction changes language models' scores on NOHARM's public cases, and whether the change depends on how much a model writes unprompted. We also sought to determine which part of the instruction carries the change and whether it is more than added length, and to assess whether the added actions are safe. This is an exploratory study on public cases scored by a locally run AI evaluator. It produces no leaderboard score and does not establish clinical safety.

Materials and Methods

Benchmark

NOHARM's cases come from real primary-care-to-specialist electronic consultations at one academic medical center.[1] The public subset, released under CC BY 4.0, has 30 base cases, three from each of ten specialties. Each appears in its original form and in ten versions that change one detail (such as age, sex, a comorbidity, a medication or a laboratory value), giving 330 related items. Each case has a rubric of candidate actions rated by specialists on a 1–9 appropriateness scale: 1,128 options in all, of which 522 are rated harmful (1–3; 79 at 1) and 198 appropriate (7–9; 23 at 9). We ran the NOHARM v2 kit unchanged except for the evaluator model (Supplementary Section S4). The organizers' held-out cases were not used.

Models and instructions

Twelve models answered the bare case (Table 1), and five of them also answered with an instruction. Freely available models may retain their inputs, so they were given public cases only. The *bare* condition sends only the case text, as the leaderboard does. The *specialist* instruction has two parts. Its *consult framing* tells the model that a primary-care colleague has asked a specialist for the best recommendations, not only an answer to the literal question. Its *plan format* asks for a one- or two-sentence assessment followed by numbered, specific recommendations (tests, drugs with doses and durations, referral specialty and urgency, conditional next steps, counseling and return precautions), leaving out nothing a careful specialist would want and including nothing unsafe. Neither part names a rubric option or sets a length.

To determine which part matters, we gave Claude Sonnet 5.5 and GPT-5.6 Terra each part alone and, as a length control, a request for at least 25 numbered recommendations that named no clinical content, which drew about as many actions as the specialist instruction. Two further instructions were used on one or two models each: a variant with eight named plan headings (*exhaustive*), and NOHARM's own priming prompt with its 500-word cap (*kit*). All instructions are given in full in Supplementary Table S1 and Section S3. They were written from the benchmark's published description of a good answer and not tuned to any case, but they were written and tried on these same public cases. Every completed arm is reported; arms stopped before completion were not analyzed.

Evaluation and outcomes

The kit's evaluator extracts the actions a plan states, matches each to the rubric, and reviews the matches. Claude Sonnet 5.5 was the primary evaluator; GPT-5.6 Terra and Nemotron 3 Ultra, a free model that wrote none of the plans, re-scored selected arms (Supplementary Section S1.3). The primary outcome was the kit's severity-weighted F1. Precision is the severity-weighted appropriateness of stated actions that match a rubric option, and recall is the severity-weighted share of appropriate options covered. We also report the share of items with at least one potentially severe error, which is a benchmark judgment rather than an adverse-event rate.

Checks on the added content

An action that matches no rubric option is never counted as harmful, and longer plans state more of these actions. We therefore recomputed F1 over all stated actions and recorded each plan's off-rubric share and length. Two model raters from other families rated every unlisted action on the specialists' 1–9 scale on a fixed 90-item sample per arm, as a screen rather than a harm rate. The author, an internal-medicine physician, then reviewed three sets of items (Supplementary Section S1.4): the unlisted actions the screens flagged, mixed with unflagged controls and blinded to model, instruction and flag (Part A); every option the specialists rated harmful that an instruction added to a model's plan, with the plan's own wording (Part B); and the instructions themselves

(Part C). Every item was answered separately: a 1–9 score for each Part A action, a verdict of harmful, equivocal or not harmful for each Part B option as the plans stated it, and a concern or none for each instruction. This item-by-item review replaces an overall verdict reported in the first version of the preprint.

Statistics

Item scores were averaged within each base case, and the 30 case means were averaged with equal weight. Every comparison is paired by case, with percentile 95% intervals from 10,000 resamples of whole cases (seed 7); cross-model statistics were recomputed on each resample. Nothing was pre-registered, and intervals are not adjusted for multiple comparisons except where Holm adjustment is stated. With 30 cases, the smallest F1 gain detectable at 80% power was 0.053 for Sonnet 5.5 and 0.102 for Terra.

Use of AI tools

Claude Code and Codex assisted with experiment code, analysis, drafting and building the physician review page; the review verdicts are the author's own. The author directed the study, checked every analysis and statement, and takes responsibility for the manuscript.

Results

The instruction improved four of five models, most for the tersest

On the bare case, F1 ranged from 0.354 (Grok 4.7) to 0.764 (Claude Sonnet 5.5) across the twelve models, and models stated a median of 3.5 to 17 actions per plan (Table 1). With an instruction, four of five models scored higher (Figure 1A), each with a Holm-adjusted bootstrap p below 0.001: Grok 4.7 by +0.279 [+0.169, +0.382], GPT-6 Astra by +0.189 [+0.120, +0.264], GPT-5.6 Terra by +0.175 [+0.108, +0.248] and Claude Sonnet 5.5 by +0.071 [+0.036, +0.110]. Muse Spark 1.3 was unchanged (+0.003 [−0.047, +0.053]). On average, F1 rose by 0.14 [0.09, 0.20]. The less a model wrote unprompted, the more it gained (Spearman -0.90 [−1.00, −0.80]). Furthermore, three instructed models (Terra, Sonnet 5.5 and Astra) scored above the highest bare score of any of the twelve models.

The gain came from recall. Terra's recall rose from 0.623 to 0.847, while its precision changed from 0.841 to 0.885. The share of items with a potentially severe error also fell under the specialist instruction (Sonnet 5.5, 0.206 to 0.155; Terra, 0.245 to 0.139). Instructing the five models halved the spread between them: the standard deviation of their scores fell from 0.17 to 0.09 (reduction 0.08 [0.03, 0.12]). Across all 26 single-call configurations, the number of actions a plan stated tracked its F1 (Spearman 0.89 [0.80, 0.92]; Figure 2).

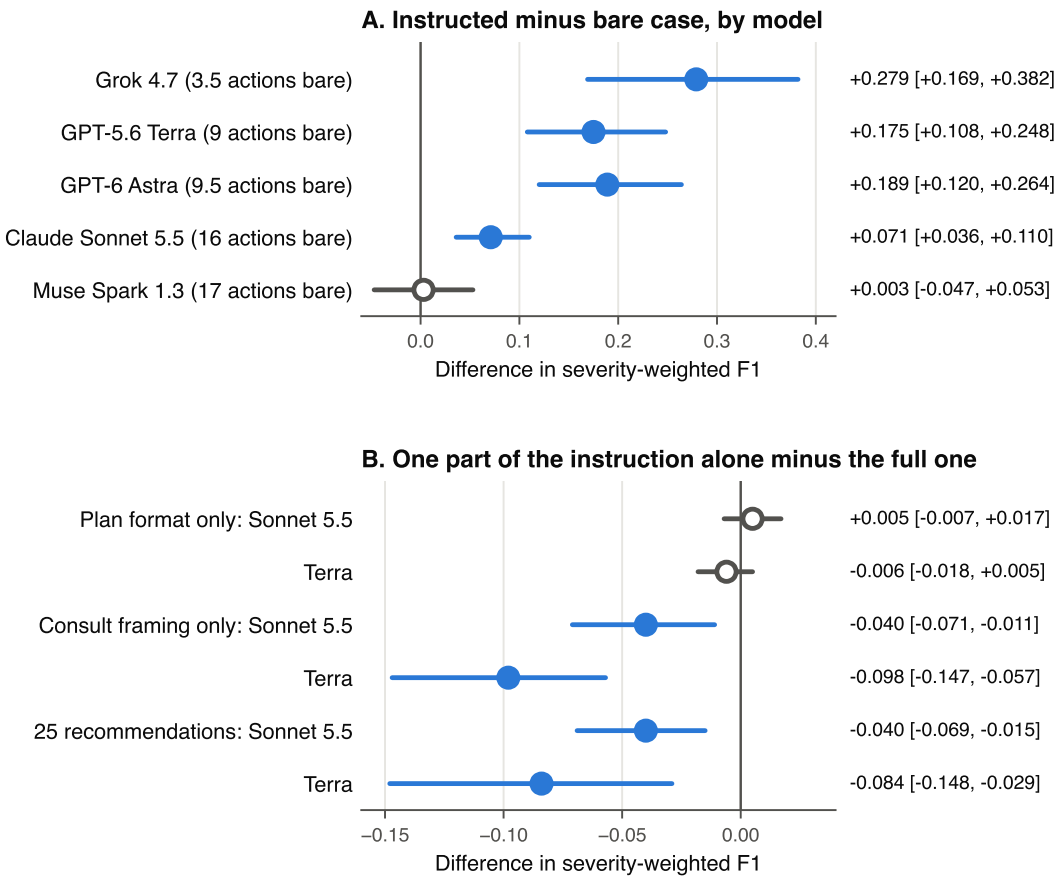


Figure 1: What the instruction changed and which part changed it. (A) Instructed minus bare-case F1 for each instructed model, ordered by how many actions the model stated unprompted. (B) For Sonnet 5.5 and Terra, the score of one part of the specialist instruction alone minus the full instruction; a difference near zero means that part reproduces the whole gain. Filled markers: interval excludes zero.

Table 1: Bare case and instructed arms, primary evaluator. F1 is severity-weighted; actions are the median number the evaluator extracted per plan; differences are paired by case with 95% intervals. Grok 4.7 and Muse Spark 1.3 received NOHARM's own priming prompt, capped at 500 words; Sonnet 5.5 and Terra the specialist instruction; Astra its exhaustive variant. Models without an instructed arm were scored bare only.

Model	Bare F1	Actions	Instruction	F1	Actions	Difference [95% CI]
Claude Sonnet 5.5	0.764	16	specialist	0.835	27	+0.071 [+0.036, +0.110]
GPT-5.6 Terra	0.678	9	specialist	0.853	21	+0.175 [+0.108, +0.248]
GPT-6 Astra	0.629	9.5	exhaustive	0.818	26	+0.189 [+0.120, +0.264]
Grok 4.7	0.354	3.5	kit, 500 words	0.633	9	+0.279 [+0.169, +0.382]
Muse Spark 1.3	0.756	17	kit, 500 words	0.759	16	+0.003 [−0.047, +0.053]
Claude Opus 5.5	0.752	15				
Claude Sonnet 5	0.599	8				
Claude Haiku 4.5	0.471	8				
GPT-5.6 Luna	0.671	10				
GPT-6.1 Sol	0.627	9				
Nemotron 3 Ultra	0.670	14				
Big Pickle	0.690	16				

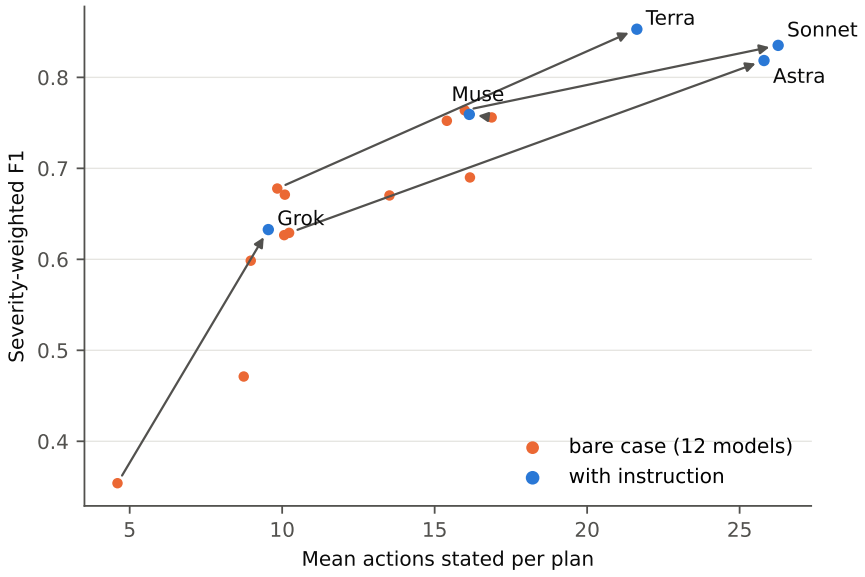


Figure 2: Median actions stated per plan against severity-weighted F1. Orange: the twelve models on the bare case; blue: the five instructed arms, each joined to its bare case by an arrow.

The plan format carried the gain, not length

When the instruction was taken apart (Table 2, Figure 1B), the plan format alone scored level with the full instruction for both models (+0.005 [−0.007, +0.017] for Sonnet 5.5, −0.006 [−0.018, +0.005] for Terra). The consult framing alone fell short of it (−0.040 [−0.071, −0.011] and −0.098 [−0.147, −0.057]). The request for 25 recommendations drew as many actions as the plan format but covered fewer indicated ones (recall 0.807 against 0.867 for Sonnet 5.5; 0.741 against 0.844 for Terra), and it scored below the full instruction (−0.040 [−0.069, −0.015] and −0.084 [−0.148, −0.029]).

Table 2: The specialist instruction taken apart, primary evaluator. F1 (all) counts every stated action in precision, including those matching no rubric option; actions and characters are medians per plan; differences are paired by case with 95% intervals.

Model	Instruction	F1	F1 (all)	Actions	Characters	vs bare	vs specialist
Sonnet 5.5	bare	0.764	0.725	16	3,202		
	framing only	0.795	0.749	19	3,988	+0.031 [+0.011, +0.052]	−0.040 [−0.071, −0.011]
	format only	0.840	0.774	28	6,464	+0.076 [+0.041, +0.115]	+0.005 [−0.007, +0.017]
	25 recommendations	0.795	0.727	29	7,407	+0.031 [+0.004, +0.058]	−0.040 [−0.069, −0.015]
	specialist	0.835	0.774	27	5,978	+0.071 [+0.036, +0.110]	
GPT-5.6 Terra	bare	0.678	0.657	9	2,206		
	framing only	0.755	0.721	12	3,703	+0.077 [+0.032, +0.127]	−0.098 [−0.147, −0.057]
	format only	0.847	0.790	22	7,038	+0.169 [+0.104, +0.241]	−0.006 [−0.018, +0.005]
	25 recommendations	0.769	0.706	24	6,628	+0.091 [+0.052, +0.135]	−0.084 [−0.148, −0.029]
	specialist	0.853	0.797	21	6,807	+0.175 [+0.108, +0.248]	

The instruction recovered follow-up and counseling

The recovered actions were those that models leave out unprompted (Table 3). Follow-up was the largest gap: coverage of appropriate follow-up options rose by +0.44 [+0.27, +0.63] for Sonnet 5.5 and +0.48 [+0.25, +0.69] for Terra, followed by counseling (+0.25 [+0.14, +0.37] and +0.21 [+0.10, +0.34]). These are the categories that NOHARM reports models and physicians alike omit.[1] In a regression of item F1 on the instruction with model and case fixed effects (3,300 item scores), the instruction's effect of +0.144 [+0.090, +0.200] fell to +0.016 [−0.023, +0.058] once the logarithm of the number of stated actions was added; the action count accounted for 89% [66, 120] of the effect. The instruction therefore works largely by making models state more actions, but which actions depends on what it asks for, since the request for 25 recommendations reached the same count and scored lower.

Safety of the added content

Actions matching no rubric option, which the headline precision ignores, made up 53% of Sonnet 5.5's bare plans and 60% of its instructed ones. When every stated action was counted in precision, each gain was smaller but persisted (Terra +0.140 [+0.071, +0.214], Sonnet 5.5

Table 3: Share of the specialists' appropriate options (rated 7–9) covered, bare → instructed, by rubric category, primary evaluator; and options the specialists rated harmful (1–3) that the plans proposed, per item. Procedure (4 appropriate options) is counted in "all"; contingency (one option) is not reported.

	Sonnet 5.5	Terra	Astra	Grok 4.7	Muse Spark
Diagnostic	0.75→0.86	0.51→0.75	0.43→0.67	0.24→0.50	0.68→0.63
Medication	0.67→0.80	0.49→0.77	0.52→0.74	0.28→0.51	0.63→0.58
Counseling	0.58→0.83	0.59→0.79	0.56→0.83	0.30→0.63	0.74→0.79
Follow-up	0.20→0.65	0.43→0.90	0.42→0.89	0.03→0.27	0.65→0.74
All appropriate	0.66→0.82	0.52→0.78	0.47→0.74	0.24→0.51	0.68→0.67
Score 9 (essential)	0.83→0.87	0.72→0.86	0.57→0.77	0.39→0.68	0.75→0.69
Harmful options proposed per item	0.29→0.42	0.11→0.12	0.10→0.14	0.24→0.31	0.25→0.24
of which severe (score 1)	0.05→0.06	0.04→0.02	0.00→0.00	0.06→0.12	0.08→0.11

+0.049 [+0.014, +0.086]). The kit's strictest variant, which counts every unlisted action as inappropriate, reversed the gain for three models (Supplementary Section S2.4). Model raters found few harmful unlisted actions in any arm. Nemotron 3 Ultra found no increase under any instruction, whereas Ling 3.1 Flash, which agreed with it only modestly (Cohen's κ 0.33), rated +0.16 [+0.04, +0.28] more of instructed Sonnet's unlisted actions per item harmful. The physician rated 17 of the 96 Part A actions harmful, 33 equivocal and 46 appropriate. All 17 harmful ratings, 9 for actions from bare plans and 8 for actions from instructed ones, fell among the 72 actions that a model rater had flagged; none fell among the 24 unflagged controls.

Against the specialists' own ratings, instructed Sonnet 5.5 proposed 0.42 harmful-rated options per item against 0.29 bare (+0.13 [+0.05, +0.23]), nearly all rated mildly or moderately harmful. For Terra, Astra, Grok and Muse Spark the change was not distinguishable from zero (Table 3). These ratings judge each option for the case in general. Read in the plans' own wording, the physician judged 13 of the 43 options that an instruction added harmful, 23 equivocal and 7 not harmful. Most plan statements of these options were conditional or offered as alternatives (123 of 227). Statements of the options judged harmful came mainly from the two models given NOHARM's capped prompt. The physician raised no concern with the specialist instruction or either of its parts, but did with the 25-recommendation minimum, the candidate menu, and an example in NOHARM's own prompt (Supplementary Section S2.4).

Evaluator checks

Re-scored by Nemotron 3 Ultra, a third model family, Terra's gain was +0.187 [+0.119, +0.259]. The primary evaluator favored an Anthropic-written plan set by +0.037 [+0.014, +0.060] relative to another model's in a paired test, which is smaller than every significant gain above. Further checks, a comparison with systems of several calls, and the reasons these scores are not comparable with the leaderboard are given in Supplementary Sections S2.5, S2.3 and S2.6.

Discussion

Principal findings

On NOHARM's public cases, a single instruction requesting a complete, structured plan improved four of five models, by up to +0.279 [+0.169, +0.382] for the tersest, and halved the spread between them. Three instructed models scored above the highest bare score of any of the twelve models, so for these models how the model was asked mattered more than which model was chosen. The plan format alone reproduced the whole gain, whereas a request for the same number of recommendations without it did not. The added actions were mainly the follow-up and counseling that models, and physicians, tend to omit. However, the added content was not free of harm: read in the plans' own wording, 13 of the 43 harmful-rated options that the instructions added were judged harmful by the physician.

What a bare score measures

These results are consistent with NOHARM's own finding that omissions dominate its errors.[1] The plan format names no clinical content, yet it recovered appropriate actions that the bare plans left out, which suggests that the models could produce these actions but did not volunteer them. A bare-case score therefore reflects two things: what a model knows and how much it chooses to say. The instruction removes much of the second difference, which explains why it narrowed the field and helped the tersest models most. The plan format appears to act as a checklist. It asks for the categories of action (follow-up, counseling, return precautions, doses) whose omission the benchmark penalizes, whereas the request for 25 recommendations, which named none of them, drew as many actions but fewer of the indicated ones. Unlike NOHARM's priming prompts, ours set no length cap. Whether the cap explains why priming left generalist models behind clinical tools in NOHARM's evaluation[1] cannot be determined from our data, because the capped prompt was given to different models here.

How firm these findings are

The direction and size of the gain for four models are the firmest result. Each gain had an adjusted p below 0.001 and persisted when every stated action was counted in precision, and Terra's gain held under a third evaluator family. The finding that the plan format carries the gain rests on two models and one wording of each part. The safety evidence is less firm and less reassuring. The rubric's harmful-rated options increased for one model, a second model rater found a small increase for the same model, and the strictest metric reverses the gain for three models. The physician also judged some added options harmful as stated, but mostly in plans from the two models given the capped prompt rather than from the model whose harmful-rated options increased, and found harmful unlisted actions in bare as well as instructed plans. All results come from public cases on which the instructions were developed, scored by an AI evaluator rather than by physicians.

Implications

For benchmark design, an unprompted protocol ranks models partly on how much they write by default. Reporting each model under a published, standard structured instruction beside its bare score, as NOHARM's own study did for priming prompts, would separate the two, as would reporting how much of a score rests on actions the rubric cannot see. Because the leaderboard admits models rather than prompts, the organizers would need to run instructed arms themselves on the held-out cases.

For clinical decision support, a structured-plan instruction is an inexpensive intervention that recovered omitted follow-up and counseling in a single call. However, it has costs. Plans under the uncapped instructions stated a median of 21 to 27 actions, which a clinician must prioritize before acting, and instructed plans included options that the physician judged harmful as stated. An instructed plan therefore needs the same scrutiny as any other. The benchmark represents advice to a primary-care clinician who reads, judges and acts on the plan. Nothing here evaluates a plan acted on without that clinician.

Limitations

The study has 30 base cases from one academic medical center, not 330 independent consultations, and the instructions were developed on these cases, so the estimates may be optimistic. Five of twelve models were instructed, with different instructions; the instruction was taken apart on two models only; and the capped and uncapped instructions were given to different models. Regeneration noise was not measured, and several planned arms were stopped and not analyzed. Every score comes from an AI evaluator, not from the organizers or from physicians. The physician review was done by the author, who also wrote the instructions, as a single reviewer; only Part A was blinded, Part B showed the panel's rating, and no independent panel or whole-plan review was used. The usability of longer plans was not measured. The public cases have been available since June 2026, so models scored later may have seen them.

Conclusion

We have found that a single structured, uncapped instruction improved NOHARM scores for four of five models and halved the spread between them, and that in the two models examined, the plan format rather than added length carried the gain. Bare-case scores of clinical completeness therefore partly rank how much models volunteer. However, instructed plans also included options that a physician judged harmful as stated. Benchmarks should report scores under a standard structured instruction beside bare scores, and check what the instruction adds, before models or clinical tools are compared. Further study is required to determine whether these gains hold on held-out cases scored by physicians.

Data availability

Every model response, evaluation, rating, the physician review materials and the analysis code are archived at `https://github.com/OJamals/noharm-prompting` (code MIT, data CC BY 4.0). The offline generators `scripts/noharm_paper_snapshot.py` and `scripts/noharm_paper_analyses.py` reproduce every number in this manuscript from that archive without model calls. The NOHARM v2 public cases are available from their authors under CC BY 4.0.

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Claude Code (Anthropic) and Codex (OpenAI) assisted with experiment code, analysis and drafting; earlier drafting used the Scientific Agent Skills workflow.[15] AI systems are not authors. The TRIPOD-LLM checklist[16] is in Supplementary Section S6.

Author contributions

Ahmad Omar Aljamal designed the study, directed the experiments and analysis, performed the physician review, and wrote and is responsible for the manuscript.

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Conflict of interest

The author declares no competing interests.

Ethics

The study reused public, de-identified benchmark data released under CC BY 4.0 and involved no patients or human participants other than the author as physician reviewer. It is not human-subjects research and was not submitted for ethics review.

References

[1] Wu D, Nateghi Haredasht F, Kumar Maharaj S, et al. First, do NOHARM: a medical safety benchmark and randomized study of physician and AI teaming on clinical consultations; 2025. Available from: `https://arxiv.org/abs/2512.01241`.

[2] Nori H, Lee YT, Zhang S, et al. Can Generalist Foundation Models Outcompete Special-Purpose Tuning? Case Study in Medicine; 2023. Available from: `https://arxiv.org/abs/2311.16452`.

[3] Ceballos-Arroyo AM, Munnangi M, Sun J, et al. Open (Clinical) LLMs are Sensitive to Instruction Phrasings. In: Proceedings of the 23rd Workshop on Biomedical Natural Language Processing (BioNLP). Bangkok, Thailand: Association for Computational Linguistics; 2024. p. 50-71. Available from: https://aclanthology.org/2024.bionlp-1.5/.

[4] Sclar M, Choi Y, Tsvetkov Y, et al. Quantifying Language Models' Sensitivity to Spurious Features in Prompt Design or: How I learned to start worrying about prompt formatting. In: The Twelfth International Conference on Learning Representations (ICLR); 2024. Available from: https://openreview.net/forum?id=RIu5lyNXjT.

[5] Mizrahi M, Kaplan G, Malkin D, et al. State of What Art? A Call for Multi-Prompt LLM Evaluation. Transactions of the Association for Computational Linguistics. 2024;12:933-49. Available from: https://aclanthology.org/2024.tacl-1.52/.

[6] Polo FM, Xu R, Weber L, et al. Efficient multi-prompt evaluation of LLMs. In: Advances in Neural Information Processing Systems. vol. 37; 2024. p. 22483-512. Available from: https://proceedings.neurips.cc/paper_files/paper/2024/hash/28236482f64a72eec43706b6f3a6c511-Abstract-Conference.html.

[7] Zheng L, Chiang WL, Sheng Y, et al. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In: Advances in Neural Information Processing Systems (Datasets and Benchmarks Track). vol. 36; 2023. p. 46595-623. Available from: https://proceedings.neurips.cc/paper_files/paper/2023/hash/91f18a1287b398d378ef22505bf41832-Abstract-Datasets_and_Benchmarks.html.

[8] Saito K, Wachi A, Wataoka K, et al. Verbosity Bias in Preference Labeling by Large Language Models; 2023. Available from: https://arxiv.org/abs/2310.10076.

[9] Dubois Y, Liang P, Hashimoto TB. Length-Controlled AlpacaEval: A Simple Debiasing of Automatic Evaluators. In: Conference on Language Modeling (COLM); 2024. Available from: https://openreview.net/forum?id=CybBmzWBX0.

[10] Panickssery A, Bowman SR, Feng S. LLM Evaluators Recognize and Favor Their Own Generations. In: Advances in Neural Information Processing Systems. vol. 37; 2024. p. 68772-802. Available from: https://proceedings.neurips.cc/paper_files/paper/2024/hash/7f1f0218e45f5414c79c0679633e47bc-Abstract-Conference.html.

[11] Wataoka K, Takahashi T, Ri R. Self-Preference Bias in LLM-as-a-Judge; 2024. Available from: https://arxiv.org/abs/2410.21819.

[12] Arora RK, Wei J, Hicks RS, et al. HealthBench: Evaluating Large Language Models Towards Improved Human Health; 2025. Available from: https://arxiv.org/abs/2505.08775.

[13] Fox S, Markham L, Lail R, et al. LLM Judges Verify Presence, Not Absence: Omission Blindness in AI Clinical Notes and What Recovers It; 2026. Available from: https://arxiv.org/abs/2608.31016.

[14] Fu HY, Shrivastava A, Moore J, et al. Absence Bench: Language Models Can't See What's Missing. In: Advances in Neural Information Processing Systems 38 (NeurIPS 2025), Datasets and Benchmarks Track. vol. 38; 2025. Available from: https://proceedings.neurips.cc/paper_files/paper/2025/hash/36b31e1bb8ecd4f4081686448e9eff2d-Abstract-Datasets_and_Benchmarks_Track.html.

[15] Kassis T, Agarwal V, He Y, et al. Scientific Agent Skills: A Library of Procedural Knowledge for Research Agents; 2026. Available from: https://arxiv.org/abs/2609.00065.

[16] Gallifant J, Afshar M, Ameen S, et al. The TRIPOD-LLM reporting guideline for studies using large lan-

guage models. Nature Medicine. 2025;31(1):60-9. Available from: https://www.nature.com/articles/s41591-024-03425-5.